When can AI discover adsorption kinetics? A discoverability map

Registered Report — Stage 1: introduction, methods and analysis plan

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Abstract

Equation-discovery methods (sparse, weak-form, errors-in-variables and symbolic regression, and universal differential equations) are proposed for learning the adsorption rate law in fixed beds. The one published demonstration on breakthrough data recovers the linear-driving-force law at a single kinetic rate and noise level, with the isotherm supplied exactly. It is not known for which regimes such discovery succeeds, or whether its failure coincides with the classical loss of practical identifiability of the mass-transfer coefficient near local equilibrium; that loss is well documented and is not claimed as new. We propose a pre-registered simulation study on data from a verified non-isothermal column solver. It varies the Damköhler number (nine levels, 10^{-1}–10^{3}), the Péclet number, noise, isotherm error, the observation model (interior profiles with or without the solid loading, or outlet only), isotherm shape (Langmuir or a MOF-303-like step) and the true law (constant k or loading-dependent $k(q)$). Discovery methods, each in its strongest configuration with documented inclusion gates, are compared on one grid with profile-likelihood identifiability of k given the true law. Three confirmatory hypotheses, each with a falsifier and pre-declared verdict words, ask whether the driving-force-to-error ratio collapses the boundary (H2a), whether discovery fails before, with or after identifiability (H2b), and whether errors-in-variables methods move the boundary (H2c); H2d, on the MOF case, is exploratory. The boundary estimator was calibrated before any discovery run, giving a minimum detectable effect of 0.25 decade. Pilots, reported separately, changed only the generator, the harness and the feasible method set.

1 Introduction

1.1 The question

A fixed-bed adsorption experiment reports concentrations and temperatures in time, usually at the column outlet. The quantity an engineer needs from it is the uptake rate law, most often written as the linear driving force (LDF), $\partial q/\partial t = k\,(q^{*}(c,T) - q)$. AI equation-discovery methods offer to find that law, not only its coefficient. Sparse regression over a library of candidate terms [1, 2], its weak-form [3–5], ensemble [6], Bayesian [7], implicit [8] and exact mixed-integer [9] variants, symbolic regression [10, 11] and neural surrogates embedded in the governing equations [12] all make that offer. This paper asks a narrower question, which can be tested:

> *For which regimes can the adsorption rate law be discovered from data a column experiment can produce, and does discoverability fail at the same boundary as the classical practical identifiability of the rate constant, or earlier?*

1.2 What is known and what is not new

The physical core of the expected failure is classical and is not claimed here. Near local equilibrium the mass-transfer coefficient is poorly determined by column data. Valocchi derived local-equilibrium validity criteria from temporal moments, in which velocity, dispersion and rate jointly decide whether local equilibrium holds [13]. Hansen and Vesselinov showed that those criteria really test whether mass-transfer-driven dispersion is negligible next to hydrodynamic dispersion [14]. Katsoulas et al. derived asymptotically that, near local equilibrium, finite mass-transfer rates enter only through an apparent dispersion coefficient [15]. In adsorption engineering, Knox et al. found that the LDF coefficient and the axial dispersion can be extracted wrongly from outlet breakthrough, and that interior measurements are needed [16]. Ward and Pini attributed about 70 % of breakthrough output variability to the isotherm parameters [17]. Taylor et al. mapped global parameter sensitivity of sorbing transport across Péclet–Damköhler regimes and found that equilibrium sorption dominates [18]. The counter-evidence is also classical. In the constant-pattern regime, a lumped rate coefficient *is* extracted from frontal breakthrough consistently with shock-layer theory [19]. "k is unidentifiable at high Damköhler number" is therefore not true without conditions. Any boundary must be conditioned on dispersion, isotherm shape, noise and isotherm error, and those are the axes of this design.

The generic failure of sparse regression on such data also has precedents. Single-trajectory identifiability of linear-in-parameter systems, the SINDy model class, depends on the trajectory's geometry; estimation sensitivity is governed by a condition number tied to the data's spatial confinement [20]. The smallest eigenvalue of the data moment matrix caps recovery for both SINDy and PySR [21]. SINDy becomes ill-conditioned on slow manifolds, where the recoverable object is the manifold itself [22]. Sparse identification shows noise- and sparsity-induced phase transitions [23]. When trajectories cross time-scale regimes, SINDy fails to find a single global model [24]. Stiffness and timescale separation break rate-law discovery in chemical kinetics [25, 26]. Collinear library functionals make the discovered model non-unique [27]. One benchmark points the other way: on attractor-filling chaotic data, SINDy performance showed no significant dependence on scale separation [28]. Breakthrough data differ from that setting. A breakthrough run is a transient single trajectory confined near the slow manifold $q \approx q^*(c, T)$, the setting of [20–22]. The design measures this confinement directly through library conditioning (§3.7).

1.3 Prior equation discovery for adsorption

The closest prior work is Santana et al. [12]. A neural network replaced the solid–fluid mass-transfer term of a fixed-bed model (a universal differential equation, UDE), was trained on outlet breakthrough curves with 5 % Gaussian noise, and was then read out by sparse regression (BIC-selected) and genetic-programming symbolic regression. For the LDF ground truth the recovered form was $p_1 + p_2 q + p_3 q^*$ with $p_2 \approx -0.22$ and $p_3 \approx 0.22$. The isotherm was supplied exactly, and one kinetic rate and one noise level were studied [12]. The same group used a universal ODE on experimental multicomponent fixed-bed data [29] and screened ten hybrid placements of neural networks in chromatography models [30]. Differentiable finite-volume hybrids now learn constitutive laws from transport experiments [31]. On the equilibrium side, neural surrogates infer the isotherm from column dynamics [32, 33]. Bootstrap confidence intervals then reveal distinct learned sorption closures with similar predictions, which is practical non-identifiability of a learned closure [34]. Symbolic regression has rediscovered isotherms [35–38]. Kinetic rate-law discovery by symbolic regression with information criteria and design of experiments exists for batch catalysis [39, 40]. Function-level identifiability of UDEs has been separated from parametric identifiability [41], and statistical tests have been used to decide whether isotherm parameters in liquid chromatography

must be state-dependent [42]. Three fetched database searches made while compiling our literature review (two OpenAlex queries and one arXiv query) found no adsorption-*kinetics* discovery study other than [12]. This supports, but does not prove, that the regime question is open.

For metal–organic framework (MOF) water sorption the gap is sharper. Water uptake on MOF-303 has been fitted with an LDF model by hand, and the specific diffusion rate falls markedly where the isotherm inflects [43]. A loading-dependent $k(q)$ at an isotherm step is an untested discovery target. A 2026 review of AI for MOF water harvesting does not mention kinetics discovery in its abstract [44].

1.4 What this study adds

Three things are new, if the hypotheses survive. The first is a quantitative discoverability boundary for the strongest available discovery methods, with a mechanism: the realised driving-force fraction against the combined error of the observations and of the isotherm the method is given. The second is a like-for-like comparison of that boundary with profile-likelihood identifiability [45] on the same grid, with the same noise, isotherm error and observation model. The third is a test of whether an errors-in-variables estimator moves the boundary. Isotherm error is an error in a regressor, not in the target, and the paper treats it so: the grid uses classical mixed least-squares–total-least-squares fitting [52, 53] inside the exact best-subset search. Errors-in-variables equation-discovery methods [46, 47] are reported on their own published examples. A negative result on any of these is publishable in the registered-report format, and the design is built to make it informative. Every null carries its minimum detectable effect, and every method that cannot be run in its strongest configuration is reported as such.

2 Hypotheses

Let δ be the realised driving-force fraction, computed from ground truth as the maximum over the observed (z, t),

$$\delta = \max_{(z,t)\ \text{observed}} |q^*(c, T) - q|/q_{\max}. \tag{1}$$

Let ε_{eff} be the combined relative error of the equilibrium the method is given (isotherm error ε), of the observations (noise σ) and, for O2, of the reconstructed state:

$$\varepsilon_{\text{eff}} = \sqrt{\sigma^2 + \varepsilon^2 + \varepsilon_{\text{rec}}^2}, \tag{2}$$

where $\varepsilon_{\text{rec}} = 0$ for O1. For O2 it is the RMS of $(\hat{q} - q)/q_{\max}$ of the mass-balance-inverted uptake (M1b, §3.4) against ground truth. It is computed once per cell from the noise-free ($\sigma = 0$) O2 observation, so it measures the inversion's structural error; the noise is already counted in σ. Then $R = \delta/\varepsilon_{\text{eff}}$, computed per cell.

H2a: the boundary is a driving-force-versus-error boundary. The probability of successful discovery, $P(\text{success})$, collapses onto one curve when plotted against R, across Da, Pe, isotherm shape and observation model (O1 and O2; O3 has no discovery method beyond M1's demonstration). The 50 % crossing lies at R^* with a spread of *less than half a decade* across conditions. *Falsifier:* an R^* spread of one decade or more across conditions. *Pre-declared words:* "*R collapses the boundary*", or "*R does not collapse the boundary; the boundary depends additionally on {named axis}*". A spread of at least half a decade but less than one decade neither confirms nor falsifies H2a and is reported in the pre-declared words "*inconclusive: R narrows the boundary to a spread of {x} decades*".

H2b: discoverability versus identifiability. For each slice (σ, ε, observation model, isotherm), we compare $\mathrm{Da}^*_{\mathrm{disc}}$, the 50 % crossing of $P(\text{success})$ for the best method, with $\mathrm{Da}^*_{\mathrm{ident}}$, the boundary beyond which k is no longer practically identifiable given the *true* law. Both are 50 % crossings from the same logistic estimator (§3.6). **H2b is confirmatory on O1 only.** There both boundaries rest on 20 replicates per level, and the 0.25-decade MDE applies. On O2 and O3, identifiability has 3 replicates on a subgrid, so those comparisons are reported descriptively, each with its own bootstrap interval, and carry no verdict word. On O1 there are three pre-declared outcomes, reported in these words:

- "*discovery fails before identifiability*": $\mathrm{Da}^*_{\mathrm{disc}} < \mathrm{Da}^*_{\mathrm{ident}}$ by at least the MDE;
- "*discovery fails with identifiability*": the absolute difference is less than the MDE;
- "*discovery outlives identifiability*": $\mathrm{Da}^*_{\mathrm{disc}} > \mathrm{Da}^*_{\mathrm{ident}}$ by at least the MDE. The method then finds the law's form where its coefficient is undetermined.

When either boundary has no crossing inside the tested range, no verdict word is issued and the range tested is reported instead (§4).

H2c: errors-in-variables methods move the boundary. The errors-in-variables (EIV) method on the grid (M5, §3.4) shifts $\mathrm{Da}^*_{\mathrm{disc}}$ by at least a factor of 2 relative to the best non-EIV method at $\varepsilon > 0$. The comparison is like for like: both methods in the strong form (M5 is strong-form only), on O1. The best non-EIV method over both forms is reported alongside, descriptively. *Falsifier:* a shift below the MDE at every $\varepsilon > 0$. *Pre-declared words:* "*EIV moves the boundary by {factor}*", or "*EIV does not move the boundary detectably (MDE {x})*". A shift at or above the MDE but below 0.30 decade (a factor of 2) is reported as "*EIV moves the boundary by {factor}, detectably but by less than the predicted factor 2*".

H2d: the MOF case (exploratory, and labelled so). For a step (Type V) isotherm with a rate that drops at the step, as reported for MOF-303 [43], is the state dependence $k(q)$ discoverable, and in which (σ, observation) cells? No directional prediction is made, and no confirmatory claim rests on H2d. Its success criterion is fixed in §3.5.

3 Methods

3.1 Ground truth

Ground truth comes from the verified finite-difference reference solver of the companion paper [48]. It is a one-dimensional, axially dispersed, non-isothermal column model, and its implementation is verified against closed-form solutions (the companion paper's L0 checks). The uptake law is one of two:

$$\text{L1:}\quad \frac{\partial q}{\partial t} = k\,(q^*(c,T) - q), \tag{3}$$

$$\text{L2:}\quad \frac{\partial q}{\partial t} = k(q)\,(q^*(c,T) - q), \qquad k(q) = k_0\Big[1 - 0.8\,s\Big(\frac{q/q_{\max} - \theta}{0.05}\Big)\Big], \tag{4}$$

where s is the logistic function and θ is the fractional loading at the isotherm's step midpoint, computed from the isotherm at the feed temperature and recorded per solve. Under L2 the rate falls to 20 % of k_0 across the step. L2 is implemented through the solver's state-dependent-rate

Table 1: Design axes (pre-registration §4.2).

axis	levels
$\mathrm{Da} = k\,t_{\mathrm{stoich}}$	9, log-spaced, 10^{-1}–10^{3} (half-decade spacing)
Pe (via the axial-dispersion multiplier)	3: $\times 0.2$, $\times 1$, $\times 5$
noise σ, i.i.d. Gaussian, relative to channel range	0, 0.5 %, 2 %, 5 %
isotherm error ε, smooth multiplicative perturbation of q^*	0, 0.5 %, 2 %, 5 %
observation model	O1 interior c, q, T; O2 interior c, T (q latent); O3 outlet $c(t), T(t)$ only
isotherm	favourable Langmuir; MOF-303-like step
true law	L1 constant k; L2 $k(q)$ (step isotherm only)

hook, whose default path is verified unchanged. Non-isothermal physics as in the companion paper is primary; an isothermal control is secondary. There are two isotherms. "Langmuir" is the companion paper's default favourable isotherm. "Step" uses the cited MOF-303 step position and capacity, with column geometry and flow held at the default so that only the isotherm changes. Ground truth is stored with a manifest that records every parameter, the stoichiometric time t_{stoich}, the horizon, and a per-solve breakthrough check (§5).

3.2 Design axes

Table 1 lists the axes. The primary Damköhler number is $\mathrm{Da} = k\,t_{\mathrm{stoich}}$, the number of kinetic time constants per front passage. The residence-time form kL/u is reported alongside but is not the axis. In this column the residence time is 0.4 s, against a front passage of about 7200 s, so $kL/u = 0.4\,\mathrm{s} \times k$. That is 4×10^{-5} to 4 over the pilot's $k = 10^{-4}$–$10\,\mathrm{s}^{-1}$, and at most about 0.04 over the targeted kinetic range, so it cannot separate the regimes this study is about (§5).

Noise and isotherm error are applied after the solve. Ground truth therefore needs only the (Da, Pe, isotherm, law) solves: $9 \times 3 \times 3 = 81$, over three (isotherm, law) pairs (Langmuir–L1, step–L1, step–L2). These 81 solves are done, at 3–7 s each at the grid's k (the generator pilot measured up to about 300 s at $k \geq 0.03\,\mathrm{s}^{-1}$). Every solve reached 95 % breakthrough, some after the horizon was extended (§5). This is ground truth, not a confirmatory result: no discovery method has seen it. Replicates are 20 independent (σ, ε) realisations per cell for M2–M6 and M9 (M8 is not run on the grid; §3.4). M6 runs on a separate isothermal control: 9 solves, the 9 Da levels $\times$ Pe $\times$ 1 $\times$ Langmuir, with the energy balance switched off (T held at the feed) and the same horizon rule. These solves are done; all reached 95 % breakthrough. M7, which passed its inclusion gate, needs a Julia search per replicate and so runs with 3 replicates on a subgrid: all Da $\times$ $\sigma \in \{0.5\,\%, 2\,\%\}$ $\times$ $\varepsilon \in \{0, 2\,\%\}$ $\times$ both isotherms $\times$ Pe $\times$ 1 $\times$ O1, the only observation model M7 can see (Table 2). M1 runs only as the minimal demonstration declared in §3.4, nothing more.

Observation model, exactly (`p2_observe.py`). All methods and the identifiability analysis read data only through one observation function, so every method sees identical corruption. O1 and O2 observe 20 equally spaced interior probes, and O3 observes the outlet; every model samples 200 times uniform over the solved horizon. Noise is i.i.d. Gaussian with standard deviation σ

times the channel's range in that cell's ground truth. The measured isotherm is $q^*_{\mathrm{meas}}(c,T) = q^*(c,T)\,[1+\varepsilon\, g(c/c_{\mathrm{in}})]$, with g a smooth random function of 4 Fourier modes on $[0,1]$, normalised to unit RMS and drawn once per replicate. Every draw is deterministic per (cell, replicate, σ, ε, observation model). The true q^* is never passed to a method.

3.3 Candidate libraries

The two libraries were fixed before the freeze to the implemented and known-answer-tested harness.

- **Lib-B (equilibrium-informed, primary):** $\{1,\ c,\ q,\ T,\ c\,q,\ q^2,\ c^2,\ q^*_{\mathrm{meas}}\}$. Including q^*_{meas} is legitimate because it is the *measured* isotherm, carrying error ε. The earlier exploratory script that motivated this study included the exact generating law, which was a defect. An earlier draft of the design specified degree-3 polynomials, $\exp(-\theta/T)$ factors and q^*q products; the tested harness uses the smaller library above. A smaller library makes discovery easier, so the boundaries measured here are *optimistic with respect to library size.*
- **Lib-A (agnostic):** Lib-B without q^*_{meas}. Under L1 the true law is not representable in Lib-A, because q^* is not polynomial, so Lib-A cannot succeed by the primary metric. It is reported as held-out fidelity only. The agnostic route to the law's form is M6 (SINDy-PI) on the isothermal control, which finds the implicit Langmuir–LDF structure without any isotherm.

3.4 Methods compared, each in its strongest configuration

The rule is that each method is run in its strongest configuration. A method that cannot be so run is reported as such and is never replaced silently by a weaker one. Implementation choices were fixed against known-answer systems before any grid cell was touched.

M1, the S1 pipeline reproduced. A UDE with a network in the uptake term, trained through the column PDE and read out by sparse and symbolic regression [12]. Its settings are at least those of [12], and stronger where those are weak. Feasibility was resolved by a pilot (§5): M1 is *not run at the declared scale.* It runs only as a labelled demonstration on a minimal subgrid: Langmuir, Pe $\times$ 1, $\sigma = 2\,\%$, $\varepsilon = 0$, O3, the 9 Da levels, one replicate, L-BFGS, at most 50 steps. The O3 comparison rests on the profile likelihood. This is a compute limit, not a finding about UDEs; on a GPU the same code is expected to be feasible.

M1b, mass-balance inversion (O2). From interior $c(z,t)$ the uptake rate follows from the gas-phase balance,

$$\frac{\partial q}{\partial t} = -\frac{\epsilon}{(1-\epsilon)\rho_p}\Big(\frac{\partial c}{\partial t} + u\frac{\partial c}{\partial z} - D_L\frac{\partial^2 c}{\partial z^2}\Big), \tag{5}$$

with q recovered by time integration from $q(z,0) = 0$. This is how an experimentalist with interior probes obtains uptake. It needs no training, and every O1 method then runs on its output, which makes it the strongest O2 competitor. Its coefficients come from the solver's own definition, so the inversion cannot drift from the forward model. Time derivatives use the same Savitzky–Golay rule as O1. Space derivatives use centred finite differences, because the probe spacing is fixed by the observation model and no smoothing window in z may be tuned to the answer. On a fresh solve that is not a grid cell, the inversion recovers q to within the 3 % of $q_{\max}$ bound recorded in the pre-registration. The test measures a maximum interior error of 2.4 % (§3.8).

M2, weak-form SINDy [3, 4]; M2b, its strong-form counterpart. The weak transform uses compactly supported test functions $\phi(t) = ((t-a)(b-t))^8$ and integration by parts, so the target needs no derivative of noisy data. The test-function support is swept over $\{31, 61, 121\}$ samples and chosen per cell by the held-out weak-system residual on every fifth probe. The law is then refitted on all probes. A support is admissible only if its training system has at least 3 equations per library term. On the known-answer test, a long support left an underdetermined weak system that fitted a wrong, dense law with zero residual, which a residual criterion would otherwise prefer. M2b is the derivative-based regression on a Savitzky–Golay $\mathrm{d}q/\mathrm{d}t$. Both forms run every sparse solver (STLSQ, M3, M4).

M3, ensemble or Bayesian SINDy [6, 7]. The inclusion probability of each term is reported. The implemented configuration is bagged sequentially thresholded least squares with an inclusion threshold of 0.6.

M4, exact best-subset selection by enumeration. This is the MIOSR objective [9], solved exactly because the library is small and so needing no commercial solver. The search is exhaustive over sparsity ≤ 4, BIC selects, and the fit is *unconstrained*. An earlier draft added sign constraints ($k > 0$, zero rate at $q = q^*$). They were dropped before the freeze because "zero rate at $q = q^*$" imposes the very coefficient ratio -1 that the success metric tests, so M4 would succeed partly by construction. Signs are judged by the metric, as for every method. BIC carries a residual-variance floor at $2 \times 10^{-3} \times \mathrm{std}(\mathrm{d}q/\mathrm{d}t)$, tied to the derivative estimator's accuracy at the grid's sampling (RMS 8×10^{-4} at 200 times). At zero noise, plain BIC kept terms of relative size 10^{-4} by fitting the estimator's own systematic error. The floor was first set at 10^{-3} from 400-sample tests, where M6 then bought a spurious q^2 term at zero noise on 200 samples. 2×10^{-3} is the smallest of $\{1, 2, 5\} \times 10^{-3}$ that passes, calibrated on the known-answer system only. These design facts were forced by known-answer tests and are declared before the freeze.

M5, errors-in-variables. *On the grid, M5 is EIV best-subset.* It runs M4's exact enumeration, but fits each support by classical mixed least-squares–total-least-squares [53], with the total-least-squares step computed by singular value decomposition [52]. Exact columns (the constant) are fitted by least squares. Error-carrying columns and the target are fitted by total least squares, after each is scaled by its *declared* error standard deviation. For the observed channels this is σ times the channel range. For q^*_{meas} it is $\varepsilon\, q^*$ plus its c-sensitivity. Products get first-order propagation. The target's standard deviation comes from the Savitzky–Golay filter's exact noise gain. Selection uses M4's floored BIC. M5 runs on the full O1 grid, with 20 replicates, in the strong form. Two known-answer tests check it: total least squares removes the attenuation bias that ordinary least squares suffers, and at zero error EIV best-subset recovers L1 with M4's support.

ODR-BINDy [46] and WENDy-IRLS [47] are reported on their own published examples only. Both learn *autonomous* ODE systems, in which every state is denoised and given its own equation. The target here is one rate equation with exogenous c and T inside a PDE, and the dominant error is in a regressor, the measured isotherm. Extending either method to that problem would be new method development. This was decided before the freeze. The ODR-BINDy Python port, on the authors' Lorenz example (500 points, 20 % noise), recovers the exact 7-term support. Its relative coefficient error is 5.0×10^{-3}, against about 2.2×10^{-3} read from the paper's heatmap, and it is about 26 times slower (2385 s). Five noise seeds, run before the freeze, settle whether that is scatter or a real gap: the mean relative coefficient error over the seeds is 0.166 against about 2.2×10^{-3} (a gap of about 74), and the exact support

is recovered in 1 of 5 seeds. The port is therefore reported, in the declared words, as "*port, accuracy below the reference*", with that gap stated; M5 on the grid is unaffected. WENDy is reported as "*NOT RUN*" if no implementation is verified on its authors' own example before the freeze.

M6, SINDy-PI [8] for rational laws. M6 is run with Lib-A and *no* measured isotherm, on the isothermal control only (§3.2), because $\exp(-\Delta H/RT)$ is not polynomial. Success is the implicit Langmuir–LDF structure $\{\mathrm{d}q,\ c\,\mathrm{d}q,\ c,\ q,\ c\,q\}$ with correct signs. Its library is $\{1,\ c,\ q,\ c\,q,\ q^2,\ c^2\}$, which is Lib-A without T, because T is constant on the isothermal control. It runs at every σ and at $\varepsilon = 0$ only, because M6 is given no isotherm and isotherm error does not apply. M6 is verified on the known-answer system, with every coefficient (hence b and k) recovered within 5 %.

M7, constrained symbolic regression with PySR [10]. Julia is installed only in an isolated environment. *Inclusion gate, written before any M7 search:* PySR (binary $+, -, \times$; maximum size 12; 40 iterations; PySR's own "best" model selection) is run on the known-answer LDF system ($k = 0.02$; 0.5 % noise on q *and* on the measured q^*_{meas}) for 3 seeds. M7 runs on its subgrid if and only if success holds in at least 2 of 3 seeds, *and* every search finishes within 30 min, *and* peak memory stays under 3 GB. Otherwise it is reported as NOT RUN, with the pilot's numbers, as "infeasible on this machine" or "failing its known-answer test", in those words. *Outcome:* all three seeds recover the planted law as $k(q^*_{\mathrm{meas}} - q)$ with k within 0.3 % of its true value, in at most 35 s and under 1 GB, so M7 is included on its subgrid. A first run of the gate recorded "infeasible on this machine"; that was our own wrapper failing — PySR rejected the library's variable names, then a non-serial deterministic setting — and the decision was withdrawn before the freeze. The gate now refuses to decide when a search raises an error, so a software failure cannot again be read as a resource limit.

M8, KAN symbolic extraction [49]. M8 follows the original KAN procedure. It is not a reimplementation of KANDy [50], whose code was not opened. It is a secondary method. *It fails its known-answer test at zero noise in every configuration tried.* The polynomial library at three sparsity penalties routes the law through q^2. Raw variables on one trajectory use q alone. Raw variables on three trajectories return $\{1, q^*, {q^*}^2\}$. This is consistent with the reported fragility of KANs on multiplicative terms [51]. M8's grid run was gated on cost. Its best configuration is raw variables with sparsity penalty 10^{-3}. Its subgrid is all Da $\times$ $\sigma \in \{0.5\,\%, 2\,\%\} \times \varepsilon \in \{0, 2\,\%\}$ $\times$ both isotherms $\times$ Pe $\times$ 1 $\times$ O1, with 3 replicates, which is 216 fits. The rule, written into the cost pilot before it ran, was that M8 runs only if one fit at the grid's size takes at most 300 s, so at most 18 h for the subgrid. *The pilot measured 1144 s, so M8 is NOT RUN on the grid.* Over the subgrid that would have been about 69 h. This is a machine limit, reported in those words. It was measured while the machine was shared with other long jobs, which the paper states. On the same known-answer data the timed fit returned only a constant, consistent with the documented failure. M8 is reported on its known-answer tests only.

M9, slow-manifold discovery [22]. M9 regresses the *state* q, not its rate, on the q-free features $\{1,\ c,\ T,\ c^2,\ q^*_{\mathrm{meas}}\}$ by exact best-subset selection. It is expected to win at high Da by returning the isotherm (plus apparent dispersion). It is reported as *what is discoverable there*, not as rate-law success. Its own success criterion is given in §3.5.

Which method can see which observation model. This was decided before the freeze; a method is never run where its input does not exist (Table 2). O1 is the optimistic ceiling. O2 and

Table 2: Method–observation applicability (pre-registration §3).

	O1 (c, q, T interior)	O2 (c, T interior)	O3 (outlet only)
M2, M2b, M3, M4, M6, M8[a] (regression on q, $\mathrm{d}q/\mathrm{d}t$)	yes	no	no
M5 EIV best-subset (mixed LS–TLS)	yes	no	no
M7 PySR	yes	no	no
M9 slow manifold	yes	no	no
M1b inversion, then every O1 method	—	yes	no
M1 UDE + sparse/symbolic readout	yes	yes	yes
profile likelihood (law known)	ODE per probe	PDE solve	PDE solve

[a] Applicable, but NOT RUN on the grid: it failed its cost gate (§3.4).

O3 are where the experimentally realistic answer lies, and O3 is reported first.

3.5 Success metric and secondary accuracy

Success (primary) means that the selected support equals the true law's support *and* every coefficient has the correct sign. For L1 in Lib-B the support must be exactly $\{q^*_{\mathrm{meas}}, q\}$, with coefficient ratio -1 within 10 %. **Accuracy (secondary)** is $|\hat{k} - k|/k \leq 10\,\%$, with $\hat{k}$ the negative of the q coefficient. $P(\text{success})$ is computed per cell over replicates. $\mathrm{Da}^*_{\mathrm{disc}}$ is obtained from a logistic fit in $\log_{10}\mathrm{Da}$, with a bootstrap confidence interval over replicates (§4). A method that raises an error in a cell is recorded as a failure with the error message, never dropped.

Success for the other targets, fixed before the freeze:

- *M9:* support $\{q^*_{\mathrm{meas}}\}$ with coefficient 1 within 5 %.
- *L2 (H2d, exploratory), operationally:* with q^*_{meas} held fixed, the fitted law f's implied rate $\hat{k}(q) = -\partial f/\partial q$ is evaluated at the observed samples and averaged in 10 equal bins of $q/q_{\max}$ over the observed range. The state dependence is *discovered* if and only if three conditions hold: the support contains q^*_{meas} and q; $\hat{k}$ decreases across the bins (Spearman $\rho < -0.8$); and $\hat{k}$ in the top bin is at most 0.6 of $\hat{k}$ in the bottom bin. The true k falls to 0.2 of its starting value. This is reported as a rate per cell, with no boundary hypothesis.
- *Lib-A fidelity:* the relative RMS error of the predicted $\mathrm{d}q/\mathrm{d}t$ (strong form) on the held-out probes (every fifth), per cell.

The best method and its selection. $\mathrm{Da}^*_{\mathrm{disc}}$ is estimated per method per slice. H2a and H2b use the method with the largest $\mathrm{Da}^*_{\mathrm{disc}}$ in the slice. Because that choice is made on the same replicates, its interval is a bootstrap that resamples replicates *jointly* across methods and re-selects the best method inside every draw, so the interval carries the selection. This is the same correction as the companion paper's selection-adjusted intervals. Every method's own $\mathrm{Da}^*_{\mathrm{disc}}$ is reported alongside.

3.6 Identifiability criterion

Identifiability is assessed by the profile likelihood of k [45] with the true law known, at the same σ, ε and observation model. k is *practically identifiable* if and only if the 95 % profile interval

(likelihood-ratio threshold 3.84) lies within $[k/2, 2k]$. In O1 the profile integrates the known-law ODE at each probe, driven by observed c and T (monotone cubic interpolation). In O2 and O3 each profile point is a full column solve, scored against the interior c, T probes (O2) or the outlet $c(t)$, $T(t)$ (O3), on a coarse 25-point k grid, with 3 replicates on the declared subgrid. Both PDE profiles are implemented and pass known-answer tests. the coarse neighbours of its interval. It makes up to 6 passes of 41 points (ODE profile) or 21 points (PDE profile), each pass narrowing to the new neighbours. At low noise the interval is narrower than one coarse step, and a single fixed refinement collapsed onto one point below the truth on the known-answer test. The profile is given the *measured* isotherm, so identifiability faces the same isotherm error as discovery.

$\mathrm{Da}^*_{\mathrm{ident}}$ is estimated exactly like $\mathrm{Da}^*_{\mathrm{disc}}$. Per replicate, k is identifiable or not under the $[k/2, 2k]$ criterion. $P(\text{identifiable})$ is fitted against $\log_{10}\mathrm{Da}$ by the same logistic estimator (§4), and $\mathrm{Da}^*_{\mathrm{ident}}$ is its 50 % crossing. The two boundaries compared in H2b are therefore measured by one estimator.

3.7 Mechanistic predictors

For each cell we record δ from ground truth, the condition number of the library Gram matrix, and its smallest eigenvalue, following [20, 21]. These test whether library conditioning, set by how closely the data hug $q = q^*$, predicts the boundary. They do not enter any verdict.

3.8 Implemented and verified components

The harness is verified on known-answer systems by a test suite (`tests/test_p2.py`). None of these systems is a grid cell. The components and their checks are:

- a synthetic LDF series obeying its law to 10^{-3} of the rate scale, and a Savitzky–Golay derivative exact to 10^{-3} on a smooth series;
- the success metric, which rejects a spurious term of size 10^{-6}, a wrong sign and a ratio error of 50 %, and accepts a 7.5 % ratio error;
- M4 exact enumeration, STLSQ and the bagged ensemble, each recovering the LDF law at zero noise (M4 with k error below 2 %);
- the weak form, recovering the law at zero noise and at 2 % noise with k error below 5 %, and beating the strong form at 2 % noise;
- the O1 runner in both forms with all three sparse solvers;
- M1b inversion to within 3 % of $q_{\max}$ on a fresh solve, and the O2 pipeline recovering the law when fed the true q. Fed the inverted q at that Da, it does not recover it, a pilot fact declared in §5;
- the differentiable column used for M1, matching the reference solver to 5×10^{-3} on the exit curve;
- the O1 profile likelihood (isothermal, and non-isothermal with temperature), the O2 interior-probe profile and the O3 outlet profile, each bracketing k within the pre-registered $[k/2, 2k]$ criterion in the kinetic regime;
- EIV best-subset: total least squares removing the attenuation bias of ordinary least squares, and recovery of L1 with M4's support at zero error;

- the L2 success metric and the Lib-A fidelity metric;
- SINDy-PI recovering the implicit Langmuir–LDF structure with no isotherm supplied, every coefficient within 5 %;
- the boundary estimator's coverage and bias (§4), and the verdict functions for H2a and H2b at the pre-declared thresholds;
- M8's known-answer test, kept as an expected failure that will break the suite if a library change makes it pass.

4 Analysis plan

4.1 Boundary estimator

$\mathrm{Da}^*_{\mathrm{disc}}$, and R^* under H2a, is the 50 % crossing of an unpenalised maximum-likelihood logistic regression of the replicate-level success outcomes on $\log_{10}\mathrm{Da}$ (or $\log_{10} R$). The confidence interval is a percentile bootstrap (2000 resamples) that resamples replicates *within* each Da level. The design is fixed, and only the stochastic replicates are resampled. No boundary is reported, and no verdict word issued, when there is no crossing inside the tested range: all successes, all failures, a fitted slope with the wrong sign, or a crossing off the grid. The tested range is reported instead. A boundary is never extrapolated off the grid.

Calibration, measured before the freeze. On 100 simulated datasets with the design's grid (9 Da levels, 20 replicates) and a known boundary, the estimator had coverage 0.94 (nominal 0.95), bias +0.012 decade, and a standard deviation of the estimate of 0.089 decade.

4.2 Minimum detectable effect

A boundary shift, or a gap between discovery and identifiability, is resolvable at about $2.8 \times 0.089 \approx 0.25$ decade (80 % power, two-sided 5 %). **The MDE is 0.25 decade**, and it is quoted with every null. It applies on O1, where discovery and identifiability both rest on 20 replicates per level. O2 and O3 identifiability has 3 replicates and is reported descriptively. The Da grid spacing is half a decade; the logistic estimator resolves the crossing well inside one grid step, which is what the measured standard deviation shows.

4.3 H2c refinement grid

A factor-2 shift is 0.30 decade, close to the MDE. H2c is therefore tested on a refined Da grid of **8 extra levels, log-uniform within ±0.5 decade of the best non-EIV method's** $\mathrm{Da}^*_{\mathrm{disc}}$ at O1, $\sigma = 2\,\%$, $\varepsilon = 2\,\%$, for each isotherm. The 8 levels lie strictly inside ±0.5 decade; the endpoints coincide with main-grid levels and are not repeated. These levels need new ground-truth solves, made after the main grid and before any EIV method is run on them. The refinement is part of the design, not a post-hoc rescue.

4.4 How each hypothesis is decided

H2a. R is computed per cell. Because σ and ε enter R itself, they are not slicing axes. A slice is one (Pe, isotherm, observation model) combination, pooling every σ, ε and Da in it. $P(\text{success})$ of the best method (§3.5) is fitted against $\log_{10} R$ by the same logistic estimator, and R^* is each slice's 50 % crossing. Only O1 and O2 enter: on O3 the sole discovery method is M1's

single-replicate demonstration, which yields no R^*. The spread is max − min of $\log_{10} R^*$ over slices, with a bootstrap interval. Slices without a crossing are listed, never imputed. The verdict words follow the thresholds in §2. When the boundary does not collapse, the axis that separates the R^* values is named.

H2b. For each O1 slice (σ, ε, isotherm), the gap is $\log_{10} \mathrm{Da}^*_{\mathrm{disc}} - \log_{10} \mathrm{Da}^*_{\mathrm{ident}}$. Here $\mathrm{Da}^*_{\mathrm{disc}}$ comes from the best method, with its selection-carrying bootstrap interval (§3.5), and $\mathrm{Da}^*_{\mathrm{ident}}$ from the same logistic estimator (§3.6). Gap ≤ -0.25 gives "*discovery fails before identifiability*"; gap $\geq +0.25$ gives "*discovery outlives identifiability*"; otherwise "*discovery fails with identifiability*". O2 and O3 gaps are reported descriptively, with intervals and no verdict words.

H2c. H2c is decided at $\sigma = 2\,\%$ on O1. The shift is

$$\log_{10} \mathrm{Da}^*_{\mathrm{disc}}(\mathrm{M5}) - \log_{10} \mathrm{Da}^*_{\mathrm{disc}}(\text{best strong-form non-EIV}),$$

with a bootstrap interval. It is estimated on the refined grid for $\varepsilon = 2\,\%$ and on the main grid for $\varepsilon \in \{0.5\,\%, 5\,\%\}$. The shift relative to the best non-EIV method over both forms is reported alongside, descriptively. The hypothesis holds at a given ε if the shift is at least $\log_{10} 2 = 0.30$ decade. A shift at or above the MDE but below 0.30 decade takes the intermediate verdict words of §2. The hypothesis is falsified if the shift is below the MDE at every $\varepsilon > 0$.

H2d. Exploratory. We report, as a rate per cell, in which (σ, observation) cells the loading-dependent rate is discovered under the L2 criterion of §3.5, with no directional prediction and no boundary hypothesis.

4.5 External anchors

First, S1 on the map. The Damköhler number of [12] will be recomputed under the definition of §3.2, and also in the residence-time form, and plotted. Our literature review estimated its residence-time value at about 0.9, which is in the kinetically controlled range. If S1 lands in our failure region, the contradiction is reported and diagnosed, not explained away. Second, real MOF-303 data (exploratory). The best method per regime is applied to published MOF-303 water-uptake kinetics [43] and to the digitised breakthrough curve the companion paper already uses. No hypothesis rests on either.

5 Pilots: what was measured and what it changed

Pilot data are excluded from every confirmatory analysis. No discovery method has been run on any grid cell. The pilots fall into two kinds. Generator pilots measured the simulator. Feasibility and verification pilots ran methods only on known-answer systems or on fresh solves that are not grid cells.

P1, generator pilot. *Measured:* for k on a log grid, the residence-time and run Damköhler numbers, grid convergence ($N_z = 400$ against 800), breakthrough and cost. *Changed:* (i) The primary Damköhler number is $k\,t_{\mathrm{stoich}}$, not kL/u. (ii) At a horizon of $1.5\,t_{\mathrm{stoich}}$ the default column reached only $c/c_{\mathrm{in}} \approx 0.75$ at the outlet. The cause is the thermal wave: the outlet was still about 9 K above feed. At $k = 0.01\,\mathrm{s}^{-1}$ the 95 % crossing was at $2.2\,t_{\mathrm{stoich}}$, and a $4\,t_{\mathrm{stoich}}$ horizon reached $c/c_{\mathrm{in}} = 1.00$. The default horizon is therefore $4\,t_{\mathrm{stoich}}$, subject to a per-solve check that the exit reaches 0.95; a solve that does not is extended, never silently truncated. (iii) The exit curve

changed by at most 0.11 % between $N_z = 400$ and 800 at every k tested (10^{-4}–$10\,\mathrm{s}^{-1}$), so $N_z = 400$ is adequate. (iv) The run Damköhler number $k\,t_{\mathrm{final}}$ spans about 1 to 10^5 over that k range. The targeted Da range 10^{-1}–10^3 therefore corresponds to $k \approx 10^{-5}$–$10^{-1}\,\mathrm{s}^{-1}$ at default physics. (v) The residence-time Damköhler number is $0.4\,\mathrm{s} \times k$, from 4×10^{-5} to 4 over the pilot's k range, which is why it is not the axis. The ground-truth solves have since been generated, at 3–7 s each. All 81 reached 95 % breakthrough within their horizon, some after extension to $8\,t_{\mathrm{stoich}}$. Generating ground truth runs no discovery method and is permitted before the freeze.

P2, M1 timing pilot. *Measured* on one cell at Da ≈ 14: the differentiable column matches the reference solver to below 5×10^{-3} on the exit curve, and $N_z = 50$ differs from $N_z = 400$ by 0.55 %. On the study CPU a forward solve takes 22 s, a gradient with respect to k 101 s, and a gradient with respect to a 2×32 network 145 s. That allows about 50 training steps in the 2 h budget, where a UDE needs hundreds to thousands. *Changed:* by the rule declared before the pilot, M1 is not run at scale (§3.4).

P3, M1b verification. *Measured* on a fresh solve that is not a grid cell: from dense exact c ($N_z = 200$), the inversion recovers q to within 3 % of $q_{\max}$. The median error was 0.013 against a median driving force of 0.18, unchanged at $N_z = 800$, so the error is not numerical diffusion. Fed the true q, the weak-form pipeline recovers the law exactly at Da ≈ 14. Fed the inverted q, it does not; it selects c^2 and T. The inversion error concentrates at the front, where the kinetic signal is. *Changed:* nothing in the methods or thresholds. The fact is declared because it bears on H2a: for O2, $\varepsilon_{\mathrm{eff}}$ must include the state-reconstruction error, and δ will be reported against it. The definition of $\varepsilon_{\mathrm{rec}}$ in §2 implements this.

P4, sampling resolution of the O1 profile. *Measured* on a fresh noise-free non-isothermal run: the profile's best k at an interior probe sits +0.61 % high with linear interpolation between 200 snapshots, +0.47 % with monotone cubic interpolation, and +0.17 % at 800 snapshots. *Changed:* monotone cubic interpolation was adopted. The bias is immaterial for the $[k/2, 2k]$ criterion and is reported with the O1/O2 results. The discovery methods face the same sampling.

P5, known-answer tests that forced design facts. These are M4's BIC residual-variance floor (and its recalibration to 2×10^{-3} at the grid's 200-sample resolution), the adaptive profile-likelihood refinement, M2's admissibility rule of at least 3 equations per library term, the documented M8 failure, and the ODR-BINDy accuracy gap (§3.4). Each was found on a known-answer system and is declared before the freeze.

P7, M8 cost gate. *Measured:* one M8 fit at the grid's size took 1144 s, against the 300 s limit written in the pilot script before it ran. The machine was shared with other long jobs at the time. *Changed:* M8 is not run on the grid (§3.4).

P6, pre-freeze revisions. Drafting this manuscript against the pre-registration and the code exposed 20 places where the two disagreed or where the pre-registration was silent. Each was fixed in the pre-registration in the open, in a dated list of pre-freeze revisions, before any discovery method ran on any grid cell. The fixes were of three kinds. Some aligned the text to the implemented and tested harness: the library, unconstrained M4, the z-derivatives and the adaptive refinement. Some defined quantities that had been named but not specified: δ, $\varepsilon_{\mathrm{eff}}$, R, the best method, $\mathrm{Da}^*_{\mathrm{ident}}$, the success criteria for M9, L2 and Lib-A, and the observation model. Others added verdict words

for the gaps between thresholds, the placement rule for the H2c levels, and the M5 statistic. One withdrew a stale MDE statement. One tightened a test: M6 now checks every coefficient to 5 %. One implemented M2's support sweep rather than dropping it. A second pass the same day did six things. It left one definition of $\mathrm{Da}^{*}_{\mathrm{ident}}$ and made H2b confirmatory on O1 only. It defined the H2a slices, $\varepsilon_{\mathrm{rec}}$, the operational L2 success rule and the Lib-A fidelity. It fixed H2c's resolution for each ε, gave the isothermal control a grid, and replaced M5's grid method with EIV best-subset. It also added the O2 profile likelihood and grid runners for every method. A third pass left one version each of the M5, M8 and L2 rules. It made H2c like for like (strong form, O1), wrote out the libraries for M6 and M9, and restricted the 0.25-decade MDE to O1.

The freeze. The two items that remained have reported: the ODR-BINDy port is reported as "port, accuracy below the reference", and M7 passes its inclusion gate. Both rules were written before the runs that decided them. The pre-registration is frozen by a commit that the grid drivers check before they will run, and no discovery method has touched a grid cell. The confirmatory grid will be run only after in-principle acceptance of this Stage-1 report. Any later change is a dated, reasoned amendment, reported as post hoc if made after results exist.

6 Timeline and compute

All computation runs on one CPU workstation without a GPU, one training job at a time, through a job queue with a completion test per job. The confirmatory grid drivers refuse to run until the freeze commit exists and the pre-registration is unchanged since it. There is no override.

- Ground truth: the 81 solves and the 9 isothermal-control solves are complete; the H2c refinement adds 8 Da levels per isotherm after the main grid.
- O1 and O2 regression methods (M2–M6, M9, and M1b followed by the O1 methods) at 20 replicates over about 2600 cells. These take seconds each. Every method has a grid runner, and every runner refuses to start before the freeze.
- M7 on its O1 subgrid, and the one-replicate M1 demonstration: about 200 searches or trainings in total.
- O2 and O3 identifiability: a full column solve per profile point, 3 replicates on the subgrid.

Order of work: the remaining pre-freeze items (§5), then the freeze, then Stage-1 review. The grid runs after in-principle acceptance, or after the internal freeze if the author decides not to wait; the Stage-2 report states which.

7 Limitations

- *Synthetic data.* The ground truth is a one-dimensional plug-flow-with-dispersion model. Non-plug flow, which can corrupt the extraction of LDF and dispersion coefficients [16], is outside the model. O3 is the realistic case and is reported first; O1 is a ceiling.
- *Competitor strength.* The claim is only as strong as the weakest strongest-configuration method. M1 is run only as a demonstration because of compute. On the grid, M5 is a classical errors-in-variables estimator, not a published EIV discovery method. ODR-BINDy and WENDy are written for autonomous systems and are reported only on their own examples. M7 passed its inclusion gate. M8 fails its known-answer test and is not run on the

grid by its cost gate, measured on a shared machine. Each is reported in those terms. Lib-B is smaller than first planned, so the boundaries are optimistic with respect to library size. Two methods recommended by our literature review are not in the competitor set: physics-constrained weak-form symbolic regression [39, 40] and partitioning by time-scale regime [24]. Their absence is a limitation, not a finding.

- *Axis choice.* Da may not be the right single axis [14, 19]. That is why Pe and isotherm shape are axes and why H2a is stated in terms of R.
- *Sampling.* Stiffness relative to the sampling interval breaks discovery in chemical kinetics [25]. The sampling here is fixed by the observation model, so at the highest Da the kinetic time is short relative to the sampling interval. This is reported, not varied.
- *Selection of the best method* per slice favours discovery, and so is conservative for the verdict "*discovery fails before identifiability*". Its interval carries the selection (§3.5).
- *Single machine.* The compute limits are those of one CPU. They decide which methods run at scale, not what the methods can do.

8 Data and code availability

The solver, harness (`p2/`), observation model, grid drivers, pilots, tests and the pre-registration with its full version history are public in the study repository, `https://github.com/abdulhamidbatayhi123/ai-mof-cof-dynamics`; the freeze commit will be identified there by its message. Ground-truth fields are regenerated deterministically from the manifest. Every corrupted observation is a deterministic function of (cell, replicate, σ, ε, observation model), so any reported number can be reproduced from the code. Pilot results files are released with the Stage-1 submission. Confirmatory results files will be released with Stage 2.

Declaration of competing interest

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Declaration of generative AI and AI-assisted technologies

During this work the author used Claude (Anthropic), an AI assistant, extensively: to write and test the method harness, the observation model, the grid drivers and the analysis code; to draft and revise the pre-registration and this Stage-1 manuscript; to retrieve and check bibliographic records; and to act as an internal reviewer. The pre-registration's version history and the test suite record what was built and when. The author reviewed and edited the content and takes full responsibility for the publication.

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